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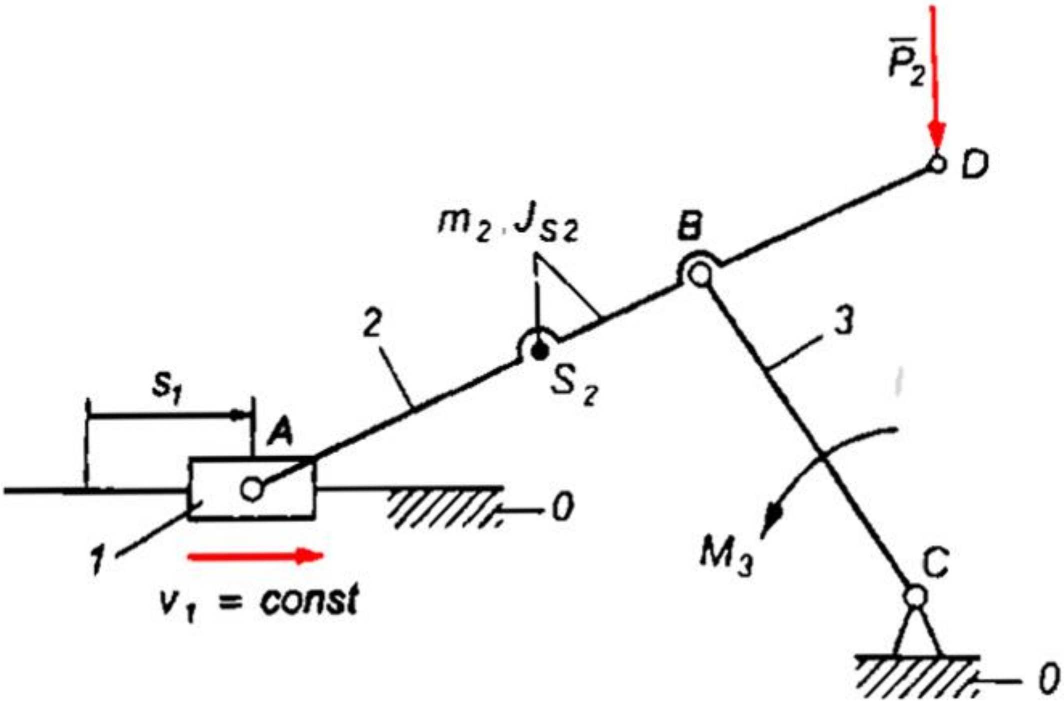
Group No: 1



Mobility The mobility w of a mechanism refers to the numbers of drives that must be added to the kinematic chain in order for the mechanism to perform precisely defined motion.

Mobility of the plane mechanism:

$$w = 6n - \sum_{i=1}^{5} i \cdot p_i$$

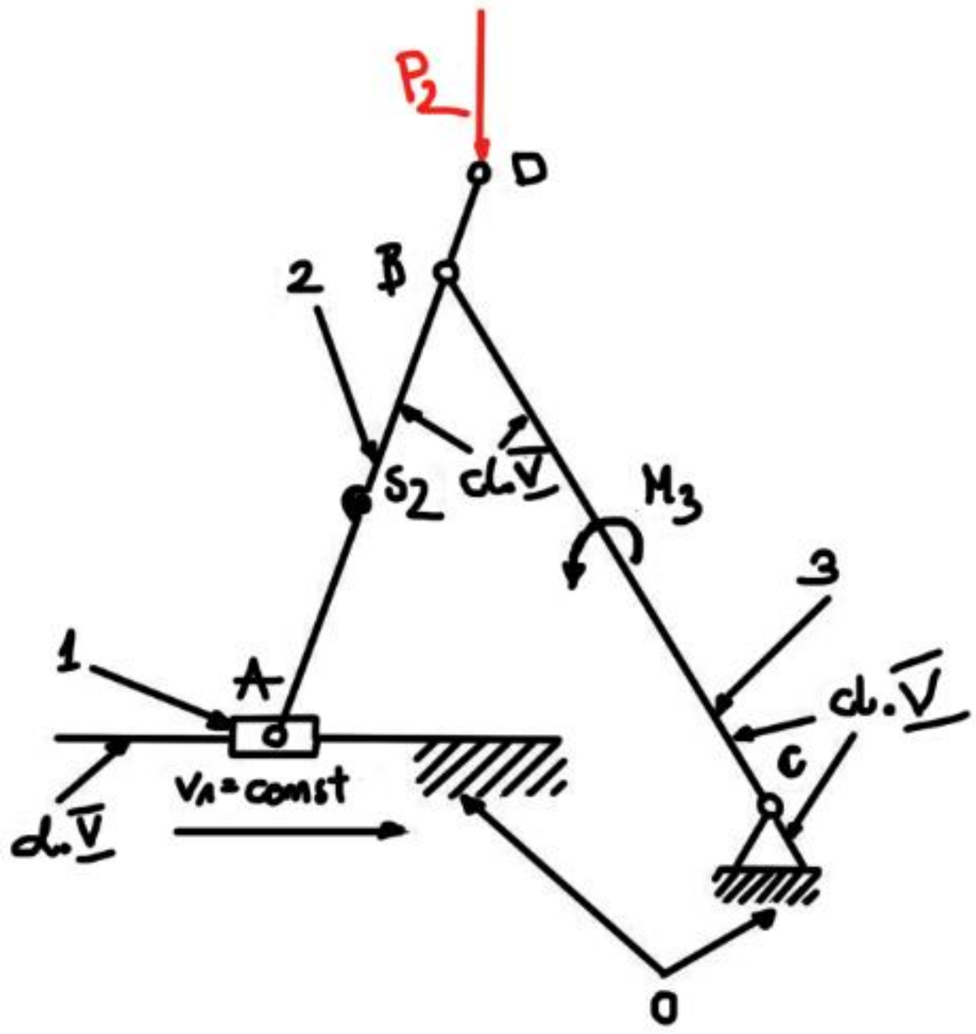
$w = 3 \cdot n - 2 \cdot p_5 - p_4$

$n = 3$

$p_5 = 4$

$p_4 = 0$

$w = 3 \cdot 3 - 2 \cdot 4 - 0 = 9 - 8 = 1$



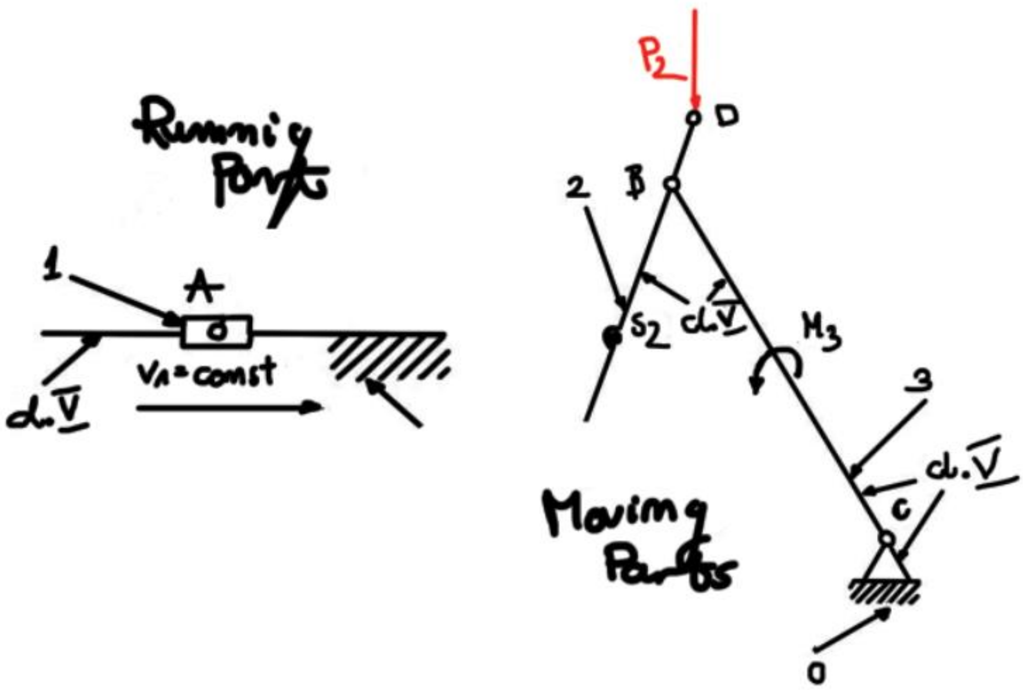
Mobility is one so mechanism requires only one driver

The mechanism is second class second form

The values chose for the project:

Value	Value in picture	Value in real life
$\|AB\|$	700mm	0.7m
$\|BC\|$	900mm	0.9m
$\|AS_3\|$	350mm	0.35m
V1	30m/s	30m/s

By choosing link 1 as a driver link, other links remain free so that it is a class 2 structural system.



Parts 2 and 3 creates a structural group of class II, whole mechanism is class II mechanism

2. Kinematic Analysis:

a) Graphoanalytical approach

acceleration of point A = $V_A = 30\ [m/s]$

changing its units $k_v = 0.3\ \left[\frac{m/s}{mm}\right]$

Length of the vector of the velocity of point A

$$(V_A) = \frac{V_A}{k_v} = \frac{30}{0.3} = 100\ mm$$

Velocity equation:

$$\overline{V_B} = \overline{V_4} + \overline{V_{B4}}$$

$(V_{B4}) = 66,56$ $\quad V_{B4} = (V_{BA}) \cdot v = 66,56 \cdot 0,3 = 19,97\ m/s$

$(V_{B4}) = 43,88\ mm$ $\quad V_B = (V_B) \cdot v = 43,88 \cdot 0,3 = 13,167\ m/s$

Then using graphical method I obtained these values:

$$\omega_2 = \frac{V_{BA}}{BA} = \frac{19{,}97}{0{,}7} = 28{,}53\,s^{-1}$$

$$\omega_3 = \frac{V_B}{BC} = \frac{131{,}167}{0{,}9} = 14{,}63\,s$$

Velocity equation of point D now instead of A, and then the same procedure with the point D now.

$$\overline{V_D} = \overline{V_A} + \overline{V_{DA}}$$

$$V_{DA} = \omega_2 \cdot AD = 28{,}53 \cdot 0{,}85 = 24{,}25$$

$$V_{DA} = 80{,}8\,mm$$

$$V_D = 36{,}65\,mm$$

$$V_D = V_D \cdot k_v = 36{,}65 \cdot 0{,}3 = 10{,}99\,m/s$$

Then the velocity of point S2

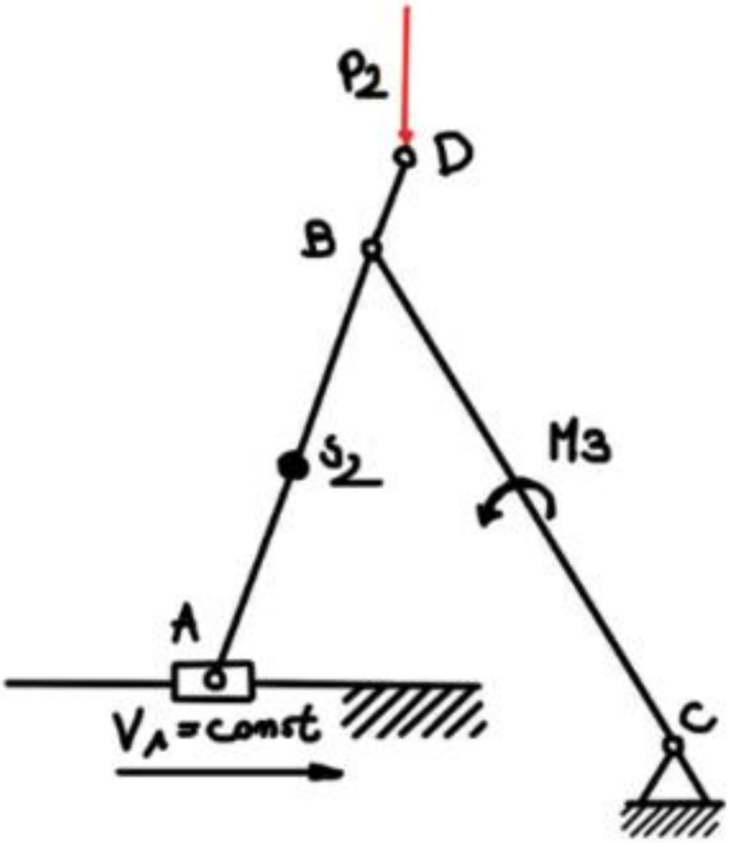
$$V_{S_2} = V_A + V_{S_{2A}}$$

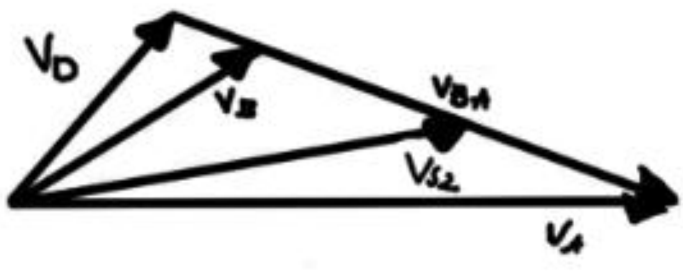
$$V_{S_{2A}} = \omega_2 \cdot |BS_2| = 28{,}53 \cdot 0{,}35 = 9{,}98\,m/s$$

$$V_{S_{2A}} = \frac{V_{S2A}}{k_v} = \frac{9{,}98}{0{,}3} = 33{,}28\,mm$$

then with the graphical approach we obtain

$$V_{S_2} = 69{,}9 \cdot 0.3 = 20{,}97\,m/s$$





Acceleration of a point A:

$$\overline{a_A} = \overline{a_1} = 0$$

Then acceleration of a point B:

$$\overline{a_B}^{n} + \overline{a_B}^{t} = \overline{a_A} + \overline{a_{BA}}^{n} + \overline{a_{BA}}^{t}$$

$$a_B^{n} = \omega_3^2 BC = 14,63^2 \cdot 0,9 = 192,63 \, m/s^2$$

$$a_{BA}^{n} = \omega_2^2 AB = 28,53^2 \cdot 0,7 = 569,77 \, m/s^2$$

The length of the acceleration point B vector:

$$|a_{BA}^{n}| = \frac{a_{BA}^{n}}{ka} = \frac{569,77}{6} \approx 94,96 \, mm$$

$$(a_{BA}^{t}) = 34,84 \, mm$$

$$a^{t}_{BA} = (a^{t}_{Ba}) \cdot ka = 206,88 \, m/s^2$$

$$(a_B^{t}) = 95,97 \, mm \qquad a_B^{t} = (a_B^{t}) \cdot ka = 95,97 \cdot 6 = 575,82 \, m/s^2$$

$$(a_B) = 101,98 \, mm$$

$$a_B = 611,88 \, m/s^2$$

Getting the results from the equations I can calculate both epsilons:

$$\varepsilon_2 = \frac{a^{t}_{BA}}{BA} = \frac{206,88}{0,7} = 295,54 \, s^{-2}$$

$$\varepsilon_3 = \frac{a^{t}_{B}}{BC} = \frac{575,82}{0,9} = 639,8 \, s^{-2}$$

Then I calculate the acceleration of the point D and vector length of the acceleration:

acceleration of the point D

$$\bar{a}_D = \bar{a}_A + \bar{a}_D{}^n + \bar{a}_D{}^t$$

$$a_D^n = \omega_2{}^2 \cdot DA = 691,86\ m/s^2$$

$$(a_D{}^n) = 115,31\ mm$$

$$a_D{}^t = \varepsilon_2 \cdot DA = 251\ m/s^2$$

$$(a_D{}^t) = 41,8\ mm$$

Acceleration of point S2:

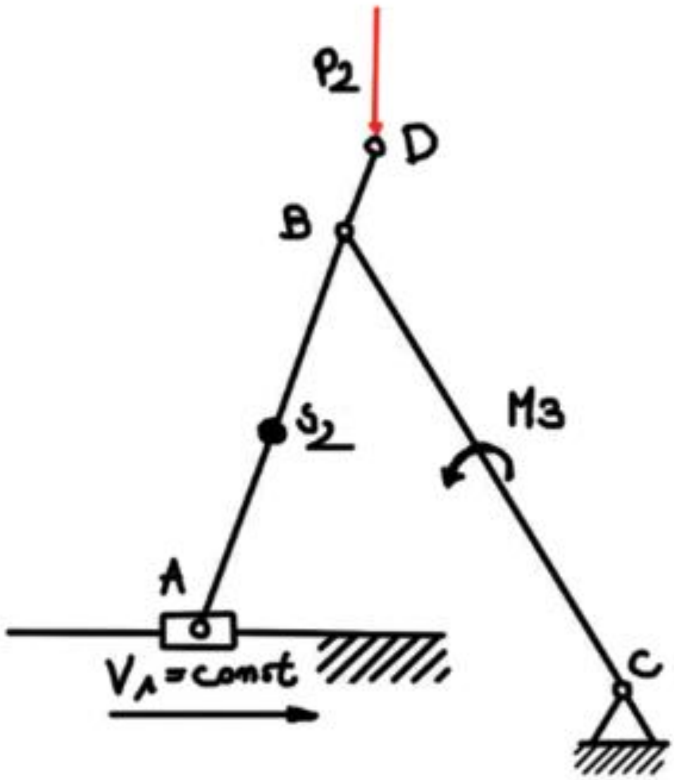
$$\{a_{S2}\} = \{\bar{a}_A\} + (a_{S2_A}^n) + (a_{S2A}{}^t)$$

$$a_{S_{2B}}^n = \omega_2{}^2 |S_2B| = 28,53^2 \cdot 0,35 = 47,48\ mm$$

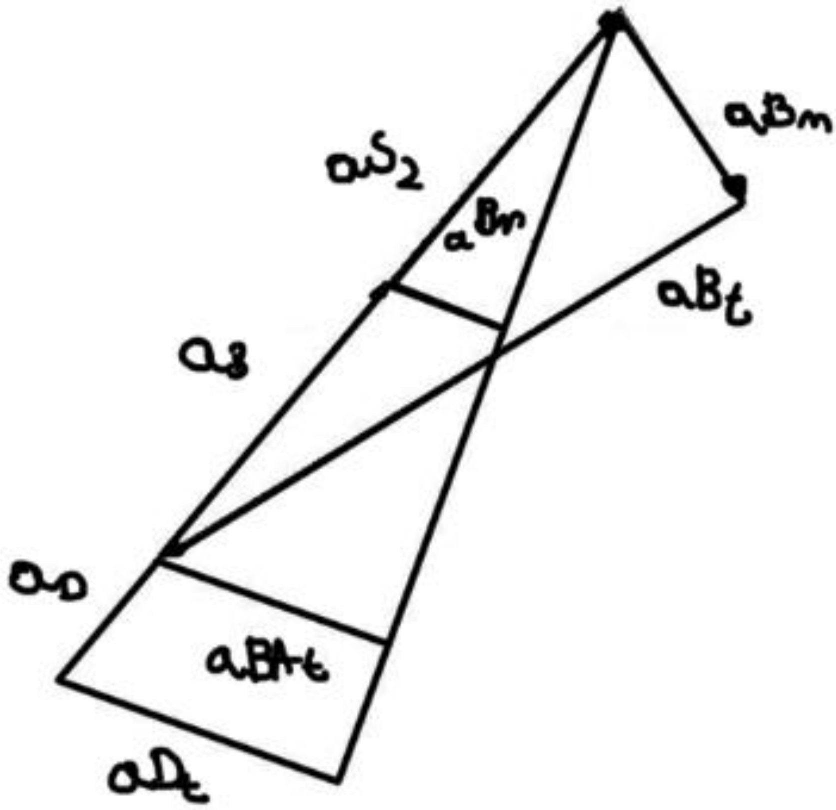
$$a_{S_{2B}}^t = \varepsilon_2 |S_2B| = 17,23\ mm$$

$$(a_{S_2}) = 50,5\ mm \qquad (a_{S_2}) \cdot ka = a_{S_2}$$

$$a_{S_2} = 303\ m/s^2$$



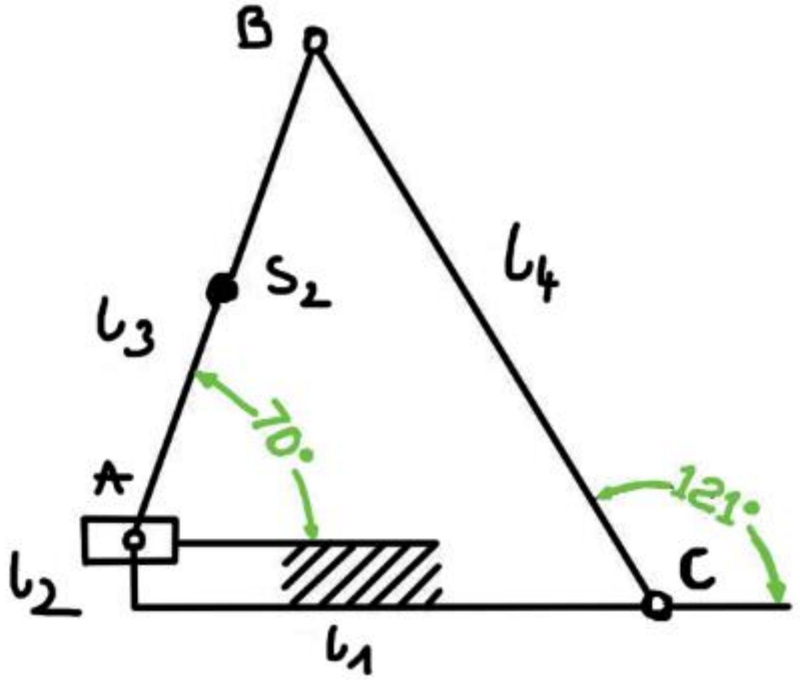
P2
D
B
S2
M3
A
C
V1 = const



aBn
aS2
aBn
aBt
aS
aD
aBAt
aDt

That is the end of the graphoanalytical approach, now I am going to the analytical approach to compare the results.

b) Analytic approach



At the beginning I set up all the values:

$l1 = 0.7\,m$

$l2 = 0.1\,m$

$l3 = 0.7\,m$

$l4 = 0.9\,m$

$\varphi_1 = 180°$ $\quad\quad \frac{dl1}{dt} = V_1 = 30\,m/s$

$\varphi_2 = 90°$

$\varphi_3 = ?$ $\quad\quad \frac{d^2 l1}{dt^2} = a_1 = 0$

$\varphi_4 = ?$

$\dot{\phi}_3 = \omega_2 \cdot \ddot{\Phi}_3 = \varepsilon_2$

Then I create both equations:

$$\overline{l_1} + \overline{l_2} + \overline{l_3} + \overline{l_4} = 0$$

$$l_1 \cos\phi_1 + l_2 \cos\phi_2 + l_3 \cos\phi_3 + l_4 \cos\phi_4 = 0$$
$$l_1 \sin\phi_1 + l_2 \sin\phi_2 + l_3 \sin\phi_3 + l_4 \sin\phi_4 = 0$$

After all calculations from equations above I obtained this:

$$\cos\phi_3 + 1{,}28 \cos\phi_4 = 1$$
$$\sin\phi_3 + 1{,}28 \sin\phi_4 = -0{,}143$$

I squared both sides:

$$-1{,}28 \cos\phi_4 + 0{,}184 \sin\phi_4 + 3{,}34 = 0$$

$$\cos\phi_4 - 0{,}143 \sin\phi_4 - 2.59 = 0$$

Later on we pulled out that the

A = - 2.59 and B = -0,143

and we get

$$(1+B^2)\cos^2\phi_4 + 2A\cos\phi_4 - (A^2 - B^2) = 0$$

After we take $w = \cos\phi_2$ we will get a squared equation

$$1{,}02w^2 - 5{,}18w - 31{,}05 = 0$$

and we will get

$w_1 = 0{,}353$ $w_2 = -0{,}548$

then 2 values

$\phi_4 = 68{,}33$

$\phi_4 = 122{,}86$

The second angle refers to the symmetrical dimensions of the axis passing through points B and D

angle ϕ_3 we calculate from the equation

$$\cos\phi_3 + 1.29\cos\phi_4 = 1 \Rightarrow \cos\phi_3 = 0,353$$

$$\phi_3 = 68,33$$

$$l_1\cos\phi_1 + l_2\cos\phi_2 + l_3\cos\phi_3 + l_4\cos\phi_4 = 0$$

after an integration

$$\frac{dl_1}{dt}\cos\phi_1 - l_3\sin\phi_1 w_2 + l_4 w3\cos\phi_4 = 0$$

where: $w_2 = \frac{d\phi_3}{dt}$; $w_3 = \frac{d\phi_4}{dt}$

In case of calculating w3, I rotate the coordinate system by the angle ϕ_3

$$\frac{dl_1}{dt}\cos(\phi_1 - \phi_3) - l_3\sin(\phi_3 - \phi_3)\cdot w_2 +$$

$$+ l_4 w_3\cos(\phi_4 - \phi_3) = 0$$

$$l_3 w_2\sin(\phi_3 - \phi_3) = 0$$

$$\Downarrow$$

we can get: $w_3 = \dfrac{\frac{dl_1}{dt}\cos(\phi_1 - \phi_3)}{l_4\cos(\phi_4 - \phi_3)}$

then $w_3 = 15,13 s^{-1}$

Symmetrically we can calculate w2 by rotating the coordinate system by ϕ_4

and it is $\omega_2 = 25,65 s^{-1}$

for calculating accelerations we take integration

and v_1 is a constant

$$\frac{dl_1}{dt} \cos\phi_1 - l_3 \sin\phi_3 \, \omega_2 + l_4 \omega_3 \cos\phi_4 = 0$$

$$\frac{d^2 l_1}{dt^2} \cos\phi_1 - l_3 \sin\phi_3 \, \varepsilon_2 - l_3 \omega_2^2 \cdot \cos\phi_3 -$$

$$- l_4 \varepsilon_3 \sin\phi_4 - l_4 \omega_3^2 \cos\phi_4 = 0$$

then we flip by the angle ϕ_3

$$\varepsilon_3 = \frac{\frac{d^2 l_1}{dt^2} \cos(\phi_1 - \phi_3) + l_3 \omega_2^2 + l_4 \omega_3^2 \cos(\phi_4 - \phi_3)}{l_4 \sin(\phi_4 - \phi_3)}$$

$$\varepsilon_3 = -636,1 s^{-2}$$

then analytically by angle ϕ_4

$$\varepsilon_2 = 293,37 s^{-2}$$

$$\omega_3 = \frac{V_B}{BC} = \frac{13,167}{0,9} \approx 14,63$$

$$V_B = \omega_3 \cdot BC = 13,617 \ m/s$$

$$V_{S_2} = 20,97$$

$$V_D = 11,3$$

$$a_{BA_n} = \omega_2^2 \cdot AB = 28,53^2 \cdot 0,7 = 569,77 \ m/s^2$$

$$a_{B_n} = \omega_3^2 \cdot BC = 200 \ m/s^2$$

$$a_{BA_t} = 293,37 \cdot 0,7 = 205,36 \ m/s^2$$

$$a_{B_t} = \varepsilon_3 \cdot BC = 512,49$$

$$\varepsilon_3 = \frac{a_t}{r} = \frac{a_{Bt}}{BC} = 640$$

$$a_{S_2} = \frac{809}{3} = 303 \ m/s^2$$

$$a_B = 611,88 \ m/s^2$$

After all calculations i compared them in the table:

Value	Graphoanalytical approach	Analytical approach m/s^2
ω_3	14,14 (rad/s^2)	15,13 (rad/s^2)
ε_3	620 (rad/s^2)	-636,1 (rad/s^2)
ε_2	290 (rad/s^2)	293,37 (rad/s^2)
VS2	20,97	21,2
VB	13,17	13,5
VD	10,99	11,3
AS2	303,0	305
aD_t	251,2	250
aD_n	691,86	695
aB_n	192,63	200
aB_t	575,82	580

The solutions match after two approaches!

1.2 Dynamic Analysis

1.2.1 – Graphoanalytical approach of dynamic analysis

I free the structural group (2,3) from constraints according to the drawing

I write down the vector equations of the equilibrium of forces acting on members 2 and 3

$$\Sigma \overline{R_{i(2)}} = \overline{R_{12}}^{n} + \overline{R_{12}}^{t} + \overline{G_2} + \overline{B_2} + R_{32} = \overline{O}$$

$$\Sigma \overline{R_{i(3)}} = \overline{R_{23}} + \overline{P_{\cancel{3}2}} + \overline{R_{O_3}} = \overline{O}$$

$$\overline{R_{23}} = -\overline{R_{\cancel{23}}32}$$

after adding both sides of an equations

$$\underline{\overline{R_{12}}}^{n} + \underline{\overline{R_{12}}}^{t} + \underline{\underline{\overline{G_2}}} + \underline{\underline{\overline{B_2}}} + \underline{\underline{\overline{P_2}}} + \underline{\overline{R_{O_3}}} = \overline{O}$$

The equation contains three unknowns, in order to take them into account in the action of the vector polygon, one of the unknowns must be included, which takes into account the balance of power moments for member 2 of the calculated due to point C, which belongs to the common kinematic pair of the group.

$$\Sigma M_{1C(2)} = 0 \Rightarrow R_{12}^{t} \cdot BC - M_2 - M_{B2} - B_2 \cdot h_2 + G_2 \cdot h_1 = 0$$

$$R_{12}^{t} = \frac{\cancel{M}M_2 + M_{B2} + B_2 \cdot h_2 - G_2 \cdot h_1}{BC}$$

$$R_{12}^{t} = -11{,}632\ N$$

$$\overline{R_{12}}^{n} + \underline{\underline{\overline{R_{12}}}}^{t} + \overline{G_2} + \overline{B_2} + \overline{P_2} + \overline{R_{O3}} = \overline{O}$$

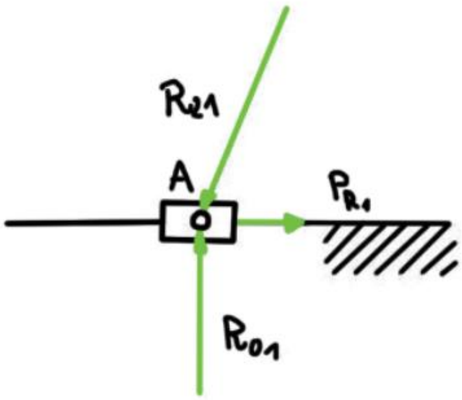
$$R_{O_3} = 24{,}553\ N$$

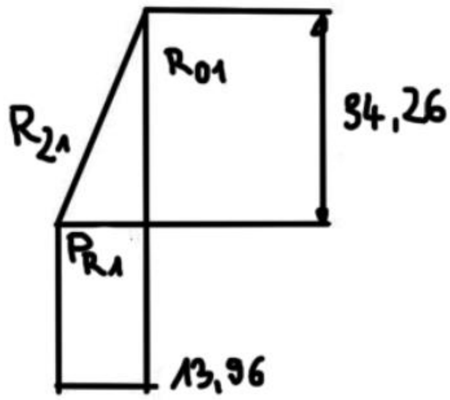
$$R_{12}^{n} = 21{,}99\ N$$

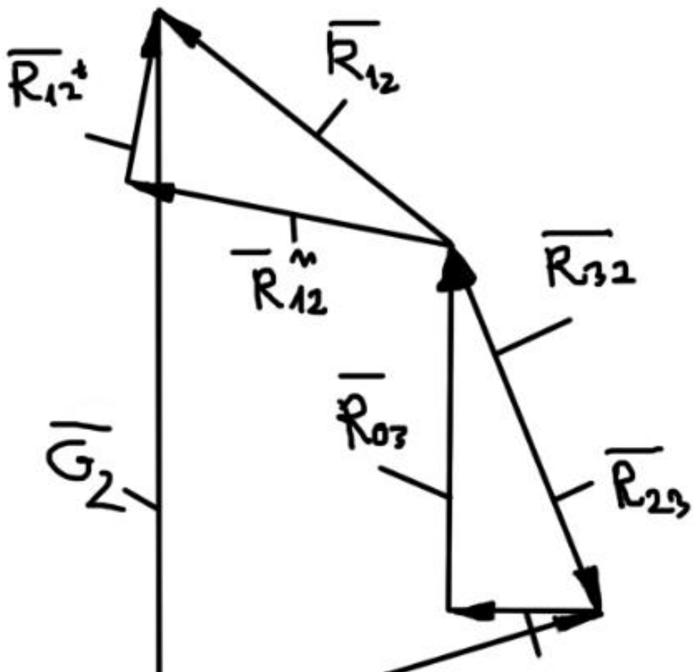
$$R_{12}^{\cancel{m}} = 24{,}877\ N$$

$$R_{23} = 26{,}511\ N$$

With the equation of moment balance about point A of armed forces on member 1 defining the equivalent moment MR1


R21
A
PR1
R01


R01
R21
94,26
PR1
13,96


R12t
R12
R12n
R32
R03
G2
R23
P3
B2

$$\bar{R}_{21} + \bar{R}_{01} = 0$$

$$\bar{R}_{21} = -\bar{R}_{01}$$

$$R_{21} = R_{01} = 24,877\,N$$

$$\sum M_{A(1)} = 0 \Rightarrow -R_{21} \cdot h_3 + M_{R_1} = 0$$

$$M_{R_1} = R_{21} \cdot h_3$$

$$h_3 = 0,23988\,m$$

$$M_{R_1} = 5,9675\,Nm$$

I obtained all the necessary values for the comparison

1.2.2 – Analytical approach

$$P_{i(1)x}: R_{01x} + R_{21x} = 0,\; P_{i(1)y}: R_{01y} + R_{21y} = 0,\; M_{i(1)A}$$

$$\Downarrow$$

$$R_{21y} \sin 60° AB + R_{21x} \cos 60° AB + M_{B2}$$

$$P_{i(2)x} \mp R_{32y} + B_2 \sin 3° + P_2 + G_2 = 0$$

Now it is solving equations with unknowns and then using Pythagoras theorem to find magnitudes

$$R_{12} = 24,872\,N$$

$$R_{23} = 26,762\,N$$

$$M_{R_1} = 5,7864\,Nm$$

1.2.3

Method of instantaneous powers:

This approach allows one to find the generalized equilibrium moment/force without determining reaction forces and check accuracy of results obtained before.

One needs all linear and angular velocities of all points and links.

$$\overline{M}_{R_1} \cdot \overline{\omega}_1 + \overline{M}_2' \overline{\omega}_2 + \overline{M}_{B2} \overline{\omega}_2 + \overline{B}_2 \overline{v}_k + \overline{G}_2 \overline{v}_k + \overline{P}_2 \overline{V}_D = \overline{0}$$

$$-M_{R_1} \cdot \omega_1 - M_2 \cdot \omega_2 - M_{B2} \cdot \omega_2 + B_2 \cdot v_k \cdot \cos\alpha + G_2 \cdot v_k \cos\beta -$$
$$- P_2 \cdot v_D = 0$$

$$M_{R_1} = \frac{-M_2 \cdot \omega_2 - M_{B2} \cdot \omega_2 + B_2 \cdot v_k \cos\alpha + G_2 \cdot v_k \cos\beta - P_2 v_D}{\omega_1}$$

$\alpha = 49°$
$\beta = 58°$

$M_{R_1} = 5{,}934\ Nm$

Comparison table of 3 different approaches of the dynamic analysis:

Value	Graphoanalytical approach	Analytical approach	Method of instantaneous powers
R12	24,877 N	24,972 N	-
R23	26,511 N	26,762 N	-
Mr1	5,9675 Nm	5,7864 Nm	5,934Nm

Methods MATCH!